

Mureșan Raul-Călin

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PROJECTS

MathSim - Bachelor's Thesis | Python, OpenCV, Tesseract OCR, Next.js [GitHub](#) | [Website](#)

- **Built** a full-stack platform that automatically ingests, digitizes, and grades a 166-page math test dataset (959 questions).
- **Engineered** a CV pipeline: contour-based segmentation, Page Sequence Voting for OCR indexing, and answer extraction via morphological dilation.
- **Optimized** image processing pipeline using multiprocessing, achieving **7.2× speedup** (reduced execution time for the full dataset from **63s to 8.7s**).
- **Developed** a FastAPI/PostgreSQL backend with JWT authentication, student/admin roles, persistent exam sessions and automatic grading, integrated with a Next.js frontend for timed quizzes and progress tracking.

SEC 10-K RAG | Python, FastAPI, React, Docker, Vercel, Ollama, Cloudflare Workers AI [GitHub](#) | [Website](#)

- **Built** and **deployed** a full-stack RAG application for SEC 10-K filings, implementing semantic chunking and cosine-similarity retrieval directly in Python, with generated answers, ranked passages and original SEC sources.
- **Packaged** immutable indexes for serverless deployment without an external vector database; supported cloud inference and local Docker/Ollama execution, with automated tests and pre-promotion CI/CD smoke checks.
- **Evaluated** on **16** held-out questions: Hit@5 of **10/12** on answerable queries; correct abstention on **4/4** unanswerable queries, with **0** false abstentions.

EXPERIENCE

Data Analyst | Robert Bosch 2025 – 2026

- **Engineered** an automated reminder system replacing manual follow-ups, reducing administrative workload for training compliance by **96%** using Power BI, Power Automate and SharePoint Lists.
- **Developed** a scalable data model using advanced Power Query (M) to map hierarchical structures across 3 management tiers.
- **Built** a real-time dashboard for training tracking, saving **~180 hours annually** by automating manual reporting tasks.

Programming Tutor | IT Junior 2024 – 2025

- **Mentored** 22 students in C, C++, HTML, CSS and Python, adapting complex curriculum to individual learning speeds and skill levels.
- **Designed** practical projects (Basic AI implementations, games, websites) to increase student engagement and technical retention.

EDUCATION

Technical University of Cluj-Napoca | B.Sc. in Computer Science 2022 – 2026 (Graduated)

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, OOP, Databases, Artificial Intelligence, Operating Systems.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, SQL, C/C++

Frameworks & Libraries: FastAPI, SQLAlchemy, React, Next.js, OpenCV

Databases & Tools: PostgreSQL, Docker, Git, Linux, pytest, GitHub Actions

Applied AI: RAG, Embeddings, Semantic Chunking, Vector Search, Ollama

Languages: Romanian (Native), English (Fluent), French (Intermediate)